Gautham Ramkumar

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Authorized to work in the US for 3 years with no sponsorship required

KEY ACHIEVEMENTS

- Placed 2nd at Revolute Hackathon, shipping a 3-model voice-to-action manipulation pipeline in one weekend

EDUCATION

Northeastern University, Boston, MA May 2026
Master of Science in Robotics (Concentration: ECE) *GPA: 3.636/4.0*

Vellore Institute of Technology, India May 2024
Bachelor of Technology in Electronics and Communications Engineering *GPA: 3.48/4.0*

TECHNICAL SKILLS

Languages	C++17, Python
Autonomy	Visual SLAM (RTAB-Map), Visual-Inertial Odometry, EKF (GNSS/IMU), GTSAM/iSAM2
Perception	Camera-LiDAR-IMU Calibration, Point Cloud Filtering (KD-tree), Stereo Vision (ZED Mini)
Hardware	NVIDIA Jetson Orin Nano, LiDAR, IMU, GNSS, SocketCAN, 6-DoF Arm
Software Practices	ROS2, ROS1 (Kalibr), Docker, Linux, Git, PyTest, GitHub Actions CI/CD, MuJoCo

PROFESSIONAL EXPERIENCE

Chicago Education Advocacy Corporation Aug 2026 – Present
Robotics AI Engineer, Chicago, IL (Remote)

- Grew automated test coverage from 44 to 145 tests across a two-job CI pipeline
- Reproduced a 70% SmolVLA task-success baseline in MuJoCo on an 8 GB GPU with pinned dependencies
- Built a dataset verifier that catches 3 silent defects (zeroed chunks, wrong state convention, no unnormalization)
- Built Backstop, a VLA failure monitor reaching 85.4% detection at 2.9% false-stop on held-out episodes

SmartInternz May 2023 – Jul 2023
Internet of Things Intern, Chennai, India

- Maintained 1 cm distance accuracy in a real-time ultrasonic sensing pipeline for automated access control

PROJECTS

Indoor 3D Mapping using RTAB-Map SLAM Sep 2025 – Oct 2025
ROS2, C++, ZED Mini | Demo

- Achieved 90% ground-truth alignment deploying visual SLAM on ZED Mini stereo hardware in ROS2
- Mapped 250 m of low-texture, GPS-denied tunnel; fixed handheld tracking loss with a rolling-chair mount

GPS-IMU-Wheel Fusion using Extended Kalman Filter Sep 2025 – Oct 2025
C++17, Eigen, ROS2, Python | Code

- Fused 200Hz IMU, 10Hz GNSS and 50Hz wheel speed in a C++17/Eigen EKF, holding position through a 462s outage
- Cut median dead-reckoning drift after 30s GPS outages 20x, from 68.83m to 3.32m, by adding wheel-speed updates

Camera-LiDAR-IMU Extrinsic and Temporal Calibration Oct 2025 – Dec 2025
C++17, Python, ROS2, GTSAM, PyTest | Code

- Rejected an unstable 150ms camera-LiDAR offset; 748 timestamp pairs gave -19.5ms within 1.5ms over 15 windows
- Tripled usable LiDAR point density per scan via a KD-tree dynamic-point filter, no segmentation network

YesChef: Voice-Controlled Robotic Sandwich Assembly Aug 2026
Python, LeRobot, Jetson Orin Nano, SocketCAN | Demo

- Added bounded retry logic around CAN control-ack timeouts, stopping crashes on the 6-DoF arm's control loop
- Fixed a leader-arm unwrap fault, rescaling the gripper so 270° of follower range spans 90° of leader travel
- Chained Whisper, Qwen2.5-3B, and Qwen2.5-VL-3B into trajectory execution on a 6-DoF arm, on-device

Stereo Visual-Inertial Odometry Calibration and Tuning Jan 2025 – Apr 2025
C++, ROS2, GTSAM/iSAM2, Kalibr, Docker, Rerun

- Improved relative pose accuracy 34.3% and cut distance drift 37.0% on an 81.47m VIO trajectory
- Exposed 9 ROS2 parameters to remove recompiles from the tuning loop